Q4: Regression

Assignment 1 | Roll number: 2024090

(a) Data loading

I loaded the training CSV into a pandas DataFrame and checked its shape, columns, data types, and target. The file contains 873 rows and 22 columns: row_id, group_id, the features F1-F19, and target. There are 291 groups, each with three rows. I kept the raw data available for exploratory analysis before applying preprocessing for modelling.

(b) Exploratory data analysis

I checked missing values, the target distribution, numerical feature-target correlations, and how the target average changed across selected inputs. This helped identify data-quality issues and relationships that a basic additive regression model might not capture.

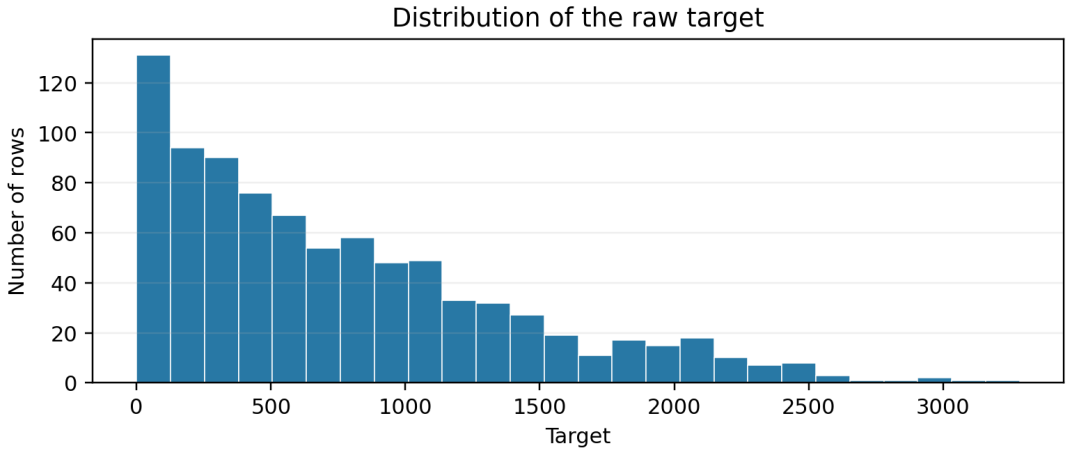


Figure 1. Distribution of the raw target.

The target is right-skewed: its mean is 755.0, its median is 595, and its maximum is 3,285. The histogram shows that most rows have lower target values while a smaller number are much larger. The numerical missing values occur in eight columns (F1, F2, F5, F8, F11, F12, F16, and F18), with about 1.3%-1.8% missing in each.

Pattern found in EDA	Decision it motivated
The target has a long right tail.	Test a fourth-root target transformation in the improved model.
The average target changes non-linearly across F10 values: it rises to around 1,440 at F10=6-7 and later falls.	Keep simple squared numerical features and test a selected interaction.
Target averages differ across the F13, F14, and F15 categories.	Use one-hot encoding rather than assigning an artificial ranking to categories.
A small number of numerical values are missing.	Use training-set medians rather than dropping incomplete rows.

These patterns motivate modelling choices, but they do not show that a feature causes the target. I used the validation results to check whether the changes were useful.

(c) Data preparation and feature engineering

(i) Choices motivated by EDA

- I excluded row_id because it only identifies a row. I used group_id to make the validation split, but did not include it as a predictor because the assignment asks about generalizing to unseen groups.
- I filled missing numerical values with medians calculated from the training data. This keeps rows that still contain useful information and is less affected by extreme values than the mean.
- I standardized the numerical columns using training-set means and standard deviations. The same fitted values are then applied to validation rows.
- I one-hot encoded F13, F14, and F15. The encoded categories are kept in a consistent column layout so that the model does not assume categories have a natural order.
- The prepared baseline contains the 16 numerical features, 14 category-indicator columns, and 16 squared standardized numerical features, giving 46 prepared inputs. The squared terms let the regression fit simple curvature.
- For the improved model, I kept those features and added selected pairwise interaction terms between F10 and the F13 category indicators. I then used Ridge Regression and a fourth-root transformation of the target. No spline features or tree-based models are used.

(ii) Suitability and use on previously unseen data

The prepared matrix has numeric values and no remaining missing numeric entries. Medians, means, standard deviations, and category levels are learned from the fitting rows and reused for validation or new rows rather than being recalculated from them. The code also checks that the number of prepared columns remains consistent when the saved preprocessing is applied to held-out rows.

If a previously unseen category appears, it has no matching indicator in the known category columns and is represented by zeros across those columns. The input format remains valid, although the model may not know the effect of that new category. These checks support reuse of the pipeline, but they do not guarantee accurate predictions for every new group.

(d) Validation strategy

I split complete groups rather than randomly splitting individual rows. Rows in the same group may share information, so placing one group in both training and validation could make validation easier than predicting a completely new group. I shuffled the unique group IDs using random seed 2024090 and held out about one fifth of the data. The same split was used to compare the baseline and improved models.

Set	Rows	Groups
Training	696	232
Validation	177	59
Total	873	291

No groups overlap between the two sets. Preprocessing is refitted on the training part of the split, then applied unchanged to the validation rows.

(e) Model training and evaluation

(i) Linear regression baseline

I used scikit-learn's LinearRegression with fit_intercept=True as the baseline. It minimizes squared prediction error and provides an interpretable starting point. It is linear in the prepared inputs; those inputs already include the squared standardized numerical features created in preprocessing.

Metric	Training	Validation
R^2	0.8165	0.7839
RMSE	275.05	286.96
MAE	206.42	216.72

(ii) Model choice and expected train-validation gap

I kept the baseline hyperparameters simple and did not tune an additional penalty for ordinary LinearRegression. Before examining the diagnostics, I expected training and validation performance to be reasonably close because this model has limited flexibility. However, similar scores alone would not prove that the model captured all the data patterns; it could still be too simple.

(f) Diagnostics

(i) What I examined

I compared training and validation metrics, plotted a learning curve using progressively larger subsets of whole training groups, and examined the validation rows with the largest absolute prediction errors. At every learning-curve point, the preprocessing was fitted using only the selected training subset, while the validation groups stayed fixed.

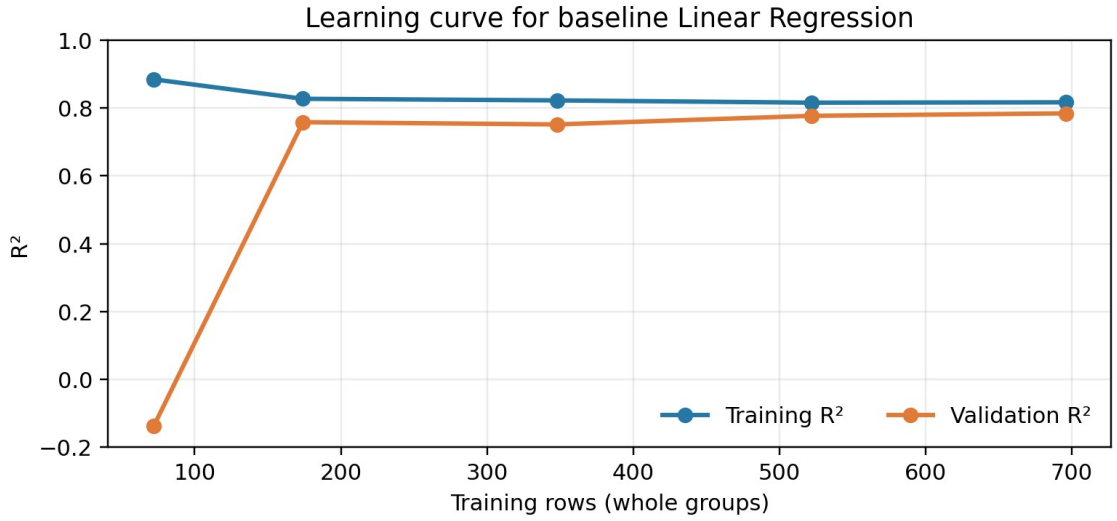


Figure 2. Learning curve for the baseline Linear Regression model.

At the full training size of 696 rows, training R^2 is 0.8165 and validation R^2 is 0.7839. The train-minus-validation R^2 gap is 0.0326.

(f) Diagnostics, continued

(ii) What the gap and learning curve show

With only 72 training rows, the training R^2 is high (0.8841), but validation R^2 is negative (-0.1371). That small training sample does not represent the full set of groups well, so the validation result is unstable. As more groups are included, validation R^2 improves into the mid-to-high 0.70s and reaches 0.7839 at full training size. The full-size train-validation gap is modest, so the baseline does not show a large overfitting gap; instead, its validation performance suggests there is still useful structure it does not capture.

(iii) Explanation of the largest errors

The baseline has errors in both directions: it underestimates some high targets and overestimates some lower targets. The table shows the five validation rows with the largest absolute residuals (residual = actual minus predicted).

Row	Group	Actual	Predicted	Residual
R0316	G004	3047	2050.5	+996.5
R0788	G208	401	1396.6	-995.6
R0335	G007	2130	1329.6	+800.4
R0843	G028	705	1413.0	-708.0
R0126	G283	2420	1740.3	+679.7

The right-tailed target distribution and the changing average target across F10 values are consistent with a model missing some structure. Only three of the 15 largest-error rows have missing predictor values, so missing data alone is unlikely to explain most of these errors.

(g) Model improvement

(i) What I changed and retrained

I kept the baseline model unchanged and used a more flexible linear model for the improvement. The improved pipeline retains the 46 prepared features and adds selected F10-by-F13 category interaction terms. These terms let the effect of F10 differ across F13 categories. I then apply StandardScaler to the combined engineered inputs and fit Ridge Regression with alpha = 0.0001.

I also used a fourth-root transformation for the target during training and reversed it when making predictions. This compresses large target values during fitting. The improvement changes the feature representation and target scale; it does not replace regression with a tree-based model.

(g) Model improvement, continued

(ii) Why the change follows from the diagnostics

The baseline learning curve and large residuals suggested that a basic additive fit was missing some relationships. Squared numerical features were already present in the prepared input, and the improved model adds a focused interaction for F10 and F13 rather than creating interactions between every possible feature. Ridge Regression adds an L2 penalty to discourage large coefficients in the expanded input. The fourth-root target transformation was tested because the target distribution has a long right tail.

(iii) Whether the improvement helped

Model	Training R^2	Validation R^2	Validation RMSE	Validation MAE
Baseline LinearRegression	0.8165	0.7839	286.96	216.72
Improved Ridge + interactions + target transform	0.8623	0.8363	249.80	170.17

Validation R^2 increased from 0.7839 to 0.8363, a gain of about 0.0523. Validation RMSE decreased by about 37.16 and MAE decreased by about 46.55. The changes helped on this group-based validation split. However, the validation R^2 is below 0.90, so this experiment does not demonstrate that the target score will be reached on the hidden test set.

(h) Generalization and limitations

(i) One way the model could fail on unseen groups

A new group may have a different relationship between the input features and target from the groups used for training. The model will still return a prediction, but it may be inaccurate for that group. An unfamiliar categorical level is another risk: its indicators are all zero in the known encoding, so the model has no learned coefficient for the new category.

(ii) One change I would try next

I would compare the baseline and improved model over several group-based validation splits, while keeping a separate outer group holdout for a final evaluation. This would show whether the improvement is consistent across different groups instead of relying on one split. I would also check whether fewer interaction terms provide similar validation performance with a simpler model.

Overall, the no-spline improved model is easier to explain than a more complex nonlinear estimator: it uses squared numerical inputs, a selected group of interactions, Ridge regularization, and a fourth-root target transformation. The validation results show an improvement over the baseline, but further group-wise validation is needed to assess generalization.